

A Stream-Fed Circumsecondary-Disk Limit Cycle in an IMRI Embedded in a Transient TDE Disk

A Working Semi-Analytic Model for Ansky-like Quasi-Periodic Eruptions

Research note for discussion

June 23, 2026

Abstract

The steadily increasing recurrence time of the quasi-periodic eruptions (QPEs) in ZTF19acnsky (‘‘Ansky’’) challenges models in which the observed clock is directly tied to a shrinking orbit. This note develops a different possibility: an intermediate-mass black hole (IMBH) embedded in a transient tidal-disruption-event (TDE) disk captures gas through its Hill sphere, forms a tidally truncated circumsecondary disk, and undergoes a thermal–viscous limit cycle. The proposed transition is *not* a generic phase change triggered by the total mass inside the Hill sphere. It is a low-to-high state transition of the bound circumsecondary disk when its surface density crosses a turning point of a multivalued thermal-equilibrium curve. We derive a one-zone vertical-structure model, identify the stress-law condition for radiation-pressure instability, estimate the critical surface density and participating mass, and connect the loading and draining times to the recurrence period, duty cycle, burst accretion rate, and luminosity. For a fiducial system with $M_{\bullet} = 10^6 M_{\odot}$, $M_2 = 10^4 M_{\odot}$, and $a = 50r_g$, the characteristic unstable-disk mass is of order 10^{-5} – $10^{-4} M_{\odot}$. Plausible loading and hot-state viscous times can then produce recurrence times of several days, duty cycles of order 0.3, and available accretion power of order $10^{44} \text{ erg s}^{-1}$. The framework remains conditional on the angular-momentum stress law, the existence of a stable advective upper branch, and the stream-fed disk geometry. We conclude with a staged numerical program in which three-dimensional hydrodynamics determines Hill-sphere capture and shock transport, while a one-dimensional radiation-disk calculation tests the limit cycle and its observable light curve.

Contents

1	Motivation and observational target	2
2	Physical architecture of the model	2
2.1	Circularization and tidal truncation	3
3	Hill-sphere supply: a two-reservoir description	4
4	Radially resolved circumsecondary-disk equations	4
5	One-zone vertical structure and local thermal equilibrium	5
5.1	Stress closure	5
6	Derivation of the low and intermediate branches	6
6.1	Cool gas-pressure branch	6
6.2	Radiation-pressure branch and instability criterion	6
6.3	The high state must be stabilized	6

7	Approximate critical surface density and mass	7
7.1	Fiducial numerical estimate	7
8	Time-dependent relaxation oscillator	8
8.1	Illustrative cycle	9
9	Secular period evolution in a transient TDE disk	9
10	Comparison with the main observational signatures	10
11	What is understood and what is not	10
12	Staged numerical and semi-analytic program	11
12.1	Layer 1: three-dimensional Hill-flow hydrodynamics	11
12.2	Layer 2: vertical-equilibrium and S-curve solver	11
12.3	Layer 3: one-dimensional time-dependent minidisk	11
12.4	Layer 4: wind and synthetic-emission model	12
13	Falsifiable conditions	12
14	Conclusions	12
A	Useful normalized scalings	13
B	Recommended minimal parameter study	13

1 Motivation and observational target

QPEs are repeating soft X-ray eruptions from galactic nuclei. Ansky is an especially stringent target because its recurrence time has increased smoothly rather than remaining constant or decreasing. The timing analysis of [Chakraborty et al. \(2026\)](#) gives

$$\dot{P} \simeq 1.7 \times 10^{-2} \text{ d d}^{-1}, \quad (1)$$

while earlier observations showed recurrence intervals of roughly 4.5–7 days. The 2025 campaign found recurrence times near 10 days, flare durations of approximately 2.5–4 days, and substantially larger flare energies than in 2024 ([Hernández-García et al., 2025](#)). A delayed UV response of about one day has subsequently been reported and provides an additional reprocessing constraint ([Guo et al., 2026](#)).

The central modeling requirements are therefore to explain, at minimum,

$$P_{\text{QPE}}, \quad \dot{P}_{\text{QPE}} > 0, \quad t_{\text{burst}}, \quad D \equiv \frac{t_{\text{burst}}}{P_{\text{QPE}}}, \quad (2)$$

plus the flare energy, soft X-ray spectrum, and multiwavelength response. The key idea explored here is that P_{QPE} is a *mass-loading clock*, not the orbital period of the secondary.

2 Physical architecture of the model

We consider a central SMBH of mass $M_{\bullet}$, an embedded secondary black hole of mass M_2 , orbital radius a , and mass ratio

$$q \equiv \frac{M_2}{M_{\bullet}} \sim 10^{-3} - 10^{-2}. \quad (3)$$

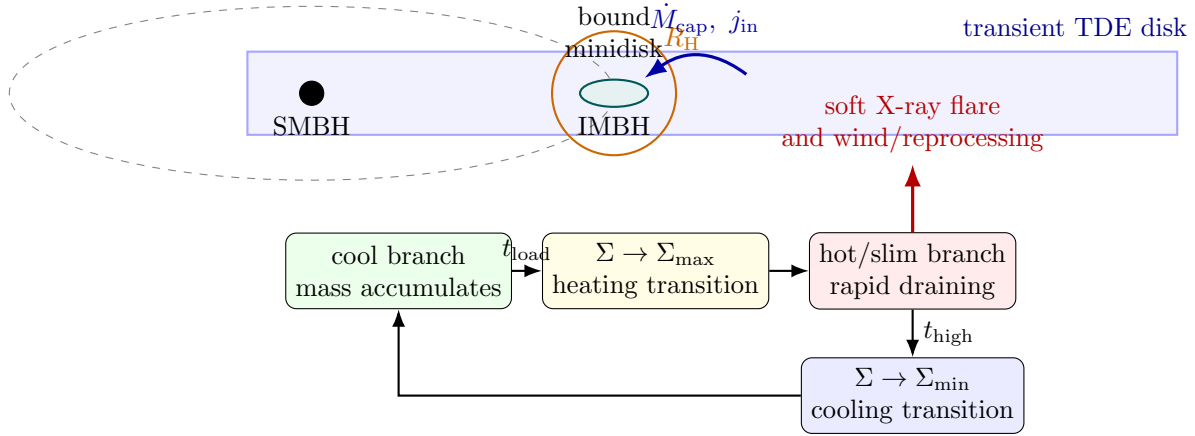


Figure 1: Schematic of the proposed stream-fed circumsecondary-disk limit cycle. The Hill flow supplies mass and angular momentum, while the burst is generated by a low-to-high state transition of the bound minidisk.

The Hill radius is

$$R_H = a \left(\frac{q}{3} \right)^{1/3}, \quad (4)$$

and the binary orbital frequency is

$$\Omega_b = \left(\frac{GM_\bullet}{a^3} \right)^{1/2}. \quad (5)$$

Equation (4) implies the useful identity

$$GM_2 = 3\Omega_b^2 R_H^3. \quad (6)$$

Gas crossing the secondary's gap does not automatically accrete. Three-dimensional simulations of analogous Hill flows show a mixture of vertical inflow, surface shocks, bound circumsecondary motion, and recycling through the Lagrange-point region (Tanigawa et al., 2012). The model therefore separates the gas into two reservoirs:

1. a recycling Hill-sphere flow with mass M_H ; and
2. a bound circumsecondary disk with mass M_d .

The proposed instability belongs to the second component.

2.1 Circularization and tidal truncation

Let the captured gas enter with characteristic specific angular momentum

$$j_{in} = \lambda \Omega_b R_H^2, \quad (7)$$

where λ encodes the three-dimensional stream geometry. Its circularization radius is

$$R_c = \frac{j_{in}^2}{GM_2} = \frac{\lambda^2}{3} R_H. \quad (8)$$

The outer disk is tidally truncated at

$$R_{out} = f_t R_H, \quad (9)$$

with $f_t < 1$. A circumsecondary disk requires

$$R_{\text{ISCO}} < R_c < R_{\text{out}}. \quad (10)$$

The values of λ , f_t , and the capture fraction are outputs of a Hill-flow simulation rather than universal constants.

3 Hill-sphere supply: a two-reservoir description

A minimal mass budget is

$$\frac{dM_H}{dt} = \dot{M}_{\text{leak}} - \dot{M}_{\text{cap}} - \frac{M_H}{t_{\text{recyc}}}, \quad (11)$$

$$\frac{dM_d}{dt} = \dot{M}_{\text{cap}} - \dot{M}_2 - \dot{M}_w, \quad (12)$$

where t_{recyc} is the Hill-flow recycling time and $\dot{M}_w$ is any wind loss from the minidisk. The external supply is written

$$\dot{M}_{\text{leak}}(t) = f_{\text{gap}}(t) \dot{M}_{\text{disk}}(a, t), \quad (13)$$

and the net bound supply is

$$\dot{M}_{\text{cap}} = f_{\text{cap}} \dot{M}_{\text{leak}}. \quad (14)$$

Both efficiencies can depend on q , H/a , viscosity, eccentricity, and the thermodynamic state of the incoming stream.

A crucial revision relative to a naive “critical Hill mass” model is that an optically thick Hill reservoir does not generically cool faster as it gains mass. For a radiation-dominated, diffusion-limited region of radius R ,

$$\tau \sim \frac{\kappa M_H}{4\pi R^2}, \quad t_{\text{cool}} \sim \frac{\tau R}{c} \sim \frac{\kappa M_H}{4\pi R c}, \quad (15)$$

so t_{cool} increases with M_H . The nonlinear transition therefore requires the structured, rotating circumsecondary disk and its coupled heating, cooling, and angular-momentum transport.

4 Radially resolved circumsecondary-disk equations

The long-term model should evolve the surface density $\Sigma(R, t)$, radial velocity v_R , and vertically integrated internal energy U . The mass equation is

$$\frac{\partial \Sigma}{\partial t} + \frac{1}{R} \frac{\partial}{\partial R} (R \Sigma v_R) = S_m(R, t) - \dot{\Sigma}_w, \quad (16)$$

where S_m describes stream deposition. The angular-momentum equation is

$$\begin{aligned} \frac{\partial(\Sigma l)}{\partial t} + \frac{1}{R} \frac{\partial}{\partial R} (R \Sigma v_R l) &= \frac{1}{R} \frac{\partial}{\partial R} (R^2 W_{R\phi}) + \Sigma \Lambda_{\text{tide}} \\ &+ S_m l_{\text{in}} - \dot{\Sigma}_w l_w, \end{aligned} \quad (17)$$

where $W_{R\phi}$ is the vertically integrated internal stress and Λ_{tide} is the tidal torque per unit mass.

The energy equation is schematically

$$\frac{\partial U}{\partial t} + \frac{1}{R} \frac{\partial}{\partial R} (R U v_R) = Q_{\text{visc}}^+ + Q_{\text{stream}}^+ + Q_{\text{tide}}^+ + Q_{\text{irr}}^+$$

$$-Q_{\text{rad}}^- - Q_{\text{adv}}^- - Q_{\text{wind}}^-. \quad (18)$$

A useful estimate of stream-impact heating is

$$Q_{\text{stream}}^+ \sim \frac{1}{2} S_m |\mathbf{v}_{\text{in}} - \mathbf{v}_d|^2. \quad (19)$$

Spiral shocks can provide significant hydrodynamic angular-momentum transport in a circumsecondary disk; in analogous circumplanetary calculations, effective values $\alpha_{\text{shock}} \sim 10^{-3} - 2 \times 10^{-2}$ have been measured (Zhu et al., 2016). This motivates separating

$$W_{R\phi} = W_{R\phi}^{\text{turb}} + W_{R\phi}^{\text{shock}}. \quad (20)$$

5 One-zone vertical structure and local thermal equilibrium

To obtain analytic insight, consider a representative annulus at radius R around the secondary, with Keplerian frequency

$$\Omega_K = \left(\frac{GM_2}{R^3} \right)^{1/2}. \quad (21)$$

The one-zone vertical relations are

$$\rho_c = \frac{\Sigma}{2H}, \quad P_c \simeq \rho_c \Omega_K^2 H^2 = \frac{1}{2} \Sigma \Omega_K^2 H, \quad (22)$$

with

$$P_c = P_{\text{gas}} + P_{\text{rad}} = \rho_c \mathcal{R} T_c + \frac{a_r T_c^4}{3}, \quad \mathcal{R} \equiv \frac{k_B}{\mu m_p}. \quad (23)$$

For an optically thick, electron-scattering annulus, the two-sided diffusion cooling rate is

$$Q_{\text{rad}}^- = \frac{16\sigma_{\text{SB}} T_c^4}{3\kappa\Sigma}. \quad (24)$$

5.1 Stress closure

The largest theoretical uncertainty is the stress law. A convenient interpolation is

$$\tau_{R\phi} = \alpha P_{\text{gas}}^\mu P_{\text{tot}}^{1-\mu}, \quad 0 \leq \mu \leq 1, \quad (25)$$

where $\mu = 0$ corresponds to total-pressure scaling and $\mu = 1$ to gas-pressure scaling. For the classical total-pressure prescription, one may write

$$\nu = \alpha \Omega_K H^2, \quad Q^+ = \frac{9}{8} \nu \Sigma \Omega_K^2, \quad \dot{M}_2 = 3\pi \nu \Sigma. \quad (26)$$

The classical radiation-pressure instability is mathematically well understood under this closure (Shakura & Sunyaev, 1973; Lightman & Eardley, 1974), but radiation-MHD simulations have shown that its nonlinear behavior depends on magnetic fluctuations, vertical transport, domain size, and stress causality (Hirose et al., 2009; Jiang et al., 2013). Thus, its applicability here must be tested rather than assumed.

6 Derivation of the low and intermediate branches

6.1 Cool gas-pressure branch

When gas pressure dominates, equations (22)–(23) give

$$H^2 \simeq \frac{\mathcal{R}T_c}{\Omega_K^2}, \quad \nu \simeq \frac{\alpha \mathcal{R}T_c}{\Omega_K}. \quad (27)$$

Then

$$Q^+ = \frac{9}{8} \alpha \mathcal{R} \Sigma \Omega_K T_c. \quad (28)$$

Equating this to equation (24) yields

$$T_c^3 = \frac{27}{128} \frac{\alpha \kappa \mathcal{R}}{\sigma_{\text{SB}}} \Sigma^2 \Omega_K, \quad (29)$$

so

$$T_c \propto \Sigma^{2/3}, \quad \dot{M}_2 \propto \Sigma^{5/3}. \quad (30)$$

At fixed Σ , $Q^+ \propto T_c$ while $Q^- \propto T_c^4$, so the branch is thermally stable.

6.2 Radiation-pressure branch and instability criterion

When radiation pressure dominates,

$$H \simeq \frac{2a_{\text{r}} T_c^4}{3\Sigma \Omega_K^2}. \quad (31)$$

For $\mu = 0$, equations (26) and (31) give

$$Q^+ \propto \frac{T_c^8}{\Sigma}, \quad Q^- \propto \frac{T_c^4}{\Sigma}. \quad (32)$$

Thus a temperature perturbation increases heating faster than cooling.

More generally, at fixed Σ in the radiation-dominated regime,

$$H \propto T_c^4, \quad P_{\text{gas}} \propto T_c^{-3}, \quad P_{\text{tot}} \propto T_c^4. \quad (33)$$

The vertically integrated heating scales as

$$Q^+ \propto H P_{\text{gas}}^{\mu} P_{\text{tot}}^{1-\mu} \propto T_c^{8-7\mu}. \quad (34)$$

Since $Q^- \propto T_c^4$, thermal instability requires

$$8 - 7\mu > 4 \quad \implies \quad \boxed{\mu < \frac{4}{7}}. \quad (35)$$

Equation (35) is a decisive condition: if the effective stress is too closely tied to gas pressure, the classical low-to-high transition disappears.

6.3 The high state must be stabilized

The radiation-pressure thin branch is the unstable connector, not necessarily the final high state. At high $\dot{M}$, radial advection, photon trapping, and winds can create a stable optically thick slim-disk branch (Abramowicz et al., 1988). The local equilibrium condition is therefore more generally

$$Q^+ = Q_{\text{rad}}^- + Q_{\text{adv}}^- + Q_{\text{wind}}^-. \quad (36)$$

Without the additional sinks in equation (36), a toy model may run away rather than produce a finite-duration burst.

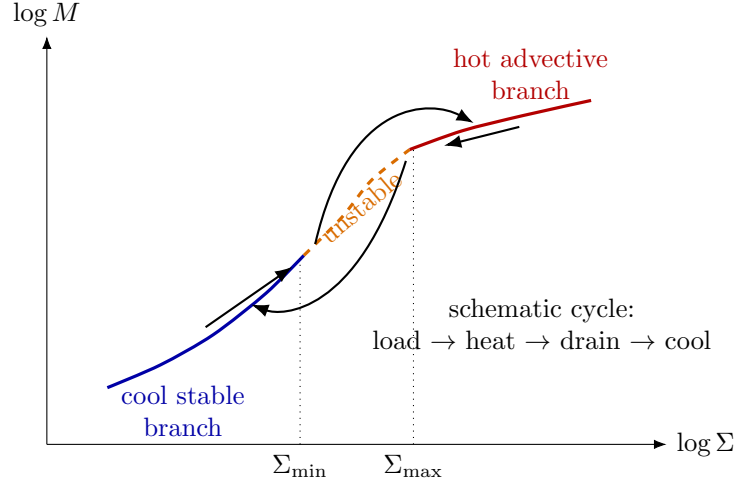


Figure 2: Schematic local equilibrium curve. The exact shape, turning points, and unstable radial interval must be calculated from the adopted stress, opacity, advection, and wind prescriptions.

7 Approximate critical surface density and mass

An order-of-magnitude estimate of the upper turning point is obtained by requiring $P_{\text{rad}} \simeq P_{\text{gas}}$ on the cool branch. Combining equation (29) with

$$\frac{a_r T_c^4}{3} = \frac{\Sigma \Omega_K}{2} \sqrt{\mathcal{R} T_c} \quad (37)$$

leads to

$$T_{\text{tr}}^4 = \frac{32 \sigma_{\text{SB}} \Omega_K}{3 \alpha_c \kappa a_{\text{r}}^2}, \quad (38)$$

and

$$\Sigma_{\text{tr}} = \left(\frac{128 \sigma_{\text{SB}}}{27 \alpha_c \kappa \mathcal{R} \Omega_K} \right)^{1/2} T_{\text{tr}}^{3/2}. \quad (39)$$

Here α_c is the effective stress parameter in the cool state. These expressions are not substitutes for a full S-curve; they provide a transparent scale for Σ_{max} .

Let R_u be the characteristic radius of the unstable annulus and let f_A denote its area relative to πR_u^2 . The critical participating mass is

$$M_{\text{crit}} \simeq f_A \pi R_u^2 \Sigma_{\text{max}}. \quad (40)$$

A radially resolved calculation should instead use

$$\Delta M_{\text{cyc}} = 2\pi \int_{R_1}^{R_2} [\Sigma_{\text{max}}(R) - \Sigma_{\text{min}}(R)] R dR. \quad (41)$$

7.1 Fiducial numerical estimate

Take

$$M_{\bullet} = 10^6 M_{\odot}, \quad M_2 = 10^4 M_{\odot}, \quad a = 50 r_g, \quad R_u = 0.3 R_{\text{H}}. \quad (42)$$

Then

$$R_{\text{H}} \simeq 1.10 \times 10^{12} \text{ cm}, \quad R_u \simeq 3.31 \times 10^{11} \text{ cm}, \quad (43)$$

with

$$\Omega_K(R_u) \simeq 6.05 \times 10^{-3} \text{ s}^{-1}. \quad (44)$$

For $\mu = 0.62$, $\kappa = 0.34 \text{ cm}^2 \text{ g}^{-1}$, and $\alpha_c = 0.01$, equations (38)–(39) give

$$T_{\text{tr}} \simeq 2.1 \times 10^6 \text{ K}, \quad \Sigma_{\text{tr}} \simeq 9.4 \times 10^5 \text{ g cm}^{-2}. \quad (45)$$

If $f_A = 1$,

$$M_{\text{crit}} \simeq 1.6 \times 10^{-4} M_{\odot}. \quad (46)$$

For $\alpha_c = 0.1$, the same estimate gives

$$M_{\text{crit}} \simeq 2.2 \times 10^{-5} M_{\odot}. \quad (47)$$

Thus a useful working range is

$$\boxed{M_{\text{crit}} \sim 10^{-5} - 10^{-4} M_{\odot}}, \quad (48)$$

subject to the uncertain unstable area f_A and stress closure.

8 Time-dependent relaxation oscillator

A reduced model can be written in terms of the minidisk mass. Let the disk leave the cool branch at M_{max} and return from the hot branch at

$$M_{\text{min}} = \zeta M_{\text{max}}, \quad 0 < \zeta < 1. \quad (49)$$

The mass processed in one cycle is

$$\Delta M = (1 - \zeta) M_{\text{max}}. \quad (50)$$

If the low-state accretion rate is small compared with the supply,

$$t_{\text{load}} \simeq \frac{\Delta M}{\dot{M}_{\text{cap}} - \dot{M}_{\text{low}}} \simeq \frac{\Delta M}{\dot{M}_{\text{cap}}}. \quad (51)$$

The hot-state draining time is approximately

$$t_{\text{high}} \sim \frac{R_u^2}{\nu_h} = \frac{1}{\alpha_h (H/R_u)_h^2 \Omega_K}. \quad (52)$$

More exactly,

$$t_{\text{high}} \simeq \frac{\Delta M}{\dot{M}_{\text{high}} - \dot{M}_{\text{cap}}}. \quad (53)$$

The observable recurrence time and duty cycle are then

$$P_{\text{QPE}} \simeq t_{\text{load}} + t_{\text{trans}} + t_{\text{high}}, \quad (54)$$

$$D \simeq \frac{t_{\text{high}}}{P_{\text{QPE}}}. \quad (55)$$

The characteristic burst rate is

$$\dot{M}_{\text{burst}} \simeq \frac{\Delta M}{t_{\text{high}}}. \quad (56)$$

8.1 Illustrative cycle

Using

$$M_{\max} = 1.6 \times 10^{-4} M_{\odot}, \quad \zeta = 0.2, \quad \dot{M}_{\text{cap}} = 10^{-2} M_{\odot} \text{ yr}^{-1}, \quad (57)$$

and

$$\alpha_h (H/R_u)_h^2 = 10^{-3}, \quad (58)$$

we obtain

$$\Delta M \simeq 1.3 \times 10^{-4} M_{\odot}, \quad (59)$$

$$t_{\text{load}} \simeq 4.8 \text{ d}, \quad (60)$$

$$t_{\text{high}} \simeq 1.9 \text{ d}, \quad (61)$$

$$P_{\text{QPE}} \simeq 6.7 \text{ d}, \quad (62)$$

$$D \simeq 0.29, \quad (63)$$

$$\dot{M}_{\text{burst}} \simeq 2.5 \times 10^{-2} M_{\odot} \text{ yr}^{-1}. \quad (64)$$

The available gravitational power is

$$L_{\text{acc}} \simeq \eta \dot{M}_{\text{burst}} c^2 \simeq 1.4 \times 10^{44} \left(\frac{\eta}{0.1} \right) \text{ erg s}^{-1}. \quad (65)$$

Equation (65) is an energy budget, not a prediction that all of the power emerges isotropically in the observed X-ray band. Photon trapping, winds, reprocessing, and beaming must be included to obtain L_X .

9 Secular period evolution in a transient TDE disk

The local supply at the IMBH orbit should be written

$$\dot{M}_{\text{cap}}(t) = f_{\text{cap}}(t) f_{\text{gap}}(t) \dot{M}_{\text{disk}}(a, t). \quad (66)$$

The raw fallback rate may initially scale approximately as $t^{-5/3}$, but viscous spreading and winds can make the local disk supply differ from fallback (Shen & Matzner, 2014). In the loading-dominated limit,

$$P_{\text{QPE}} \simeq \frac{\Delta M_{\text{cyc}}}{\dot{M}_{\text{cap}}}. \quad (67)$$

Hence

$$\frac{\dot{P}_{\text{QPE}}}{P_{\text{QPE}}} \simeq \frac{d \ln \Delta M_{\text{cyc}}}{dt} - \frac{d \ln \dot{M}_{\text{cap}}}{dt}. \quad (68)$$

If ΔM_{cyc} changes slowly and $\dot{M}_{\text{cap}} \propto t^{-5/3}$,

$$P_{\text{QPE}} \propto t^{5/3}, \quad \frac{\dot{P}}{P} = \frac{5}{3t}. \quad (69)$$

This gives the required sign without orbital expansion. However, equation (68) is the more important result: changes in the unstable annulus, stress, disk thickness, or capture fraction can modify both the magnitude of $\dot{P}$ and the flare energy.

The observed increase in both recurrence time and flare energy from 2024 to 2025 is not automatically reproduced by a constant-threshold, declining-supply model. It may require an increasing ΔM_{cyc} , for example because the unstable annulus expands, Σ_{max} rises, or the hot-state draining time lengthens as the external disk evolves. This is a central quantitative test rather than a solved feature.

Observable	Model connection	Present status
Positive $\dot{P}$	A declining captured supply increases the loading time; equation (68) does not require orbital outspiral.	Natural sign; magnitude depends on TDE-disk evolution.
$P_{\text{QPE}} \sim \text{several--10 d}$	P is the sum of loading and hot-state draining times.	Reproduced by the fiducial example.
$t_{\text{burst}} \sim 2.5\text{--}4 \text{ d}$ and large duty cycle	The flare duration is the viscous/depletion time of the hot branch, not a dynamical free-fall time.	Plausible for $\alpha_h(H/R)^2 \sim 10^{-3}$; must be solved radially.
High flare energy and L_X	A $10^3\text{--}10^4 M_\odot$ secondary makes the required accretion power much less extreme than for a stellar-mass BH.	Energy budget is plausible; X-ray efficiency is uncertain.
Soft X-ray spectrum	Super-Eddington inner flow and/or wind photosphere can reprocess harder emission.	Requires spectral and wind modeling.
Longer, more energetic 2025 flares	Could reflect growth of the unstable area or threshold mass as the external disk changes.	Important unresolved constraint.
Regular but not perfectly periodic bursts	A deterministic S-curve gives a relaxation clock; noisy stream supply and shocks introduce jitter.	Qualitatively plausible; needs time-dependent calculation.
Delayed UV response	Irradiation or diffusion through an outflow/reprocessing layer can lag the X-ray high state.	Compatible in principle, not yet predicted.

Table 1: Qualitative comparison of the model with Ansky-like observational constraints.

10 Comparison with the main observational signatures

11 What is understood and what is not

The proposed mechanism combines a well-known mathematical instability with a poorly explored astrophysical environment.

Relatively well understood. Given a specified stress law, opacity, and advective closure, one-dimensional disk equations can be used to construct local thermal equilibria, identify stable and unstable branches, and integrate limit cycles. Radiation-pressure and slim-disk branches have a substantial theoretical literature (Shakura & Sunyaev, 1973; Lightman & Eardley, 1974; Abramowicz et al., 1988).

Not established for this system. It is unknown whether a tidally truncated, stream-fed IMBH minidisk has the necessary stress law, unstable radial interval, and stable high branch. Magnetic support can narrow or remove the unstable region in TDE disks (Kaur et al., 2023); conversely, radiation-MHD calculations have found both long-lived equilibria and runaway behavior under different conditions (Hirose et al., 2009; Jiang et al., 2013). Existing QPE disk-instability models demonstrate that limit cycles can reproduce some light-curve properties, but they do not establish this particular circumsecondary geometry (Pan et al., 2022).

Decisive theoretical question. Does a stream-fed, tidally truncated circum-IMBH disk possess two stable branches separated by an unstable branch over a radial interval that fits inside R_{out} ? If not, the model fails before any light-curve fitting is attempted.

12 Staged numerical and semi-analytic program

A full radiation-MHD calculation would be ideal but is not required for the first decisive tests. A computationally economical program has four layers.

12.1 Layer 1: three-dimensional Hill-flow hydrodynamics

The hydro simulation should determine the supply and geometry, not claim to derive the radiation-pressure limit cycle. Recommended outputs are

$$\dot{M}_{\text{cap}}(t), \quad j_{\text{in}}(t), \quad s_{\text{in}}(t), \quad R_c, \quad t_{\text{recyc}}, \quad Q_{\text{stream}}^+, \quad \alpha_{\text{shock}}. \quad (70)$$

A practical setup is a local three-dimensional corotating domain spanning several R_H , with the secondary potential, tidal and Coriolis forces, controlled outer-disk inflow, and a sink much smaller than R_c . The parameter scan should include

$$q = 10^{-3} - 10^{-2}, \quad H/a = 0.03 - 0.1, \quad (71)$$

plus multiple thermodynamic closures. Constant- β cooling can be used to bracket how capture depends on entropy loss, but t_{cool} is then an imposed input. The measured quantities are the capture response and stream thermodynamics, not a first-principles radiation instability.

The shock stress can be estimated from

$$W_{R\phi}^{\text{shock}} = \int \rho \delta v_R \delta v_\phi dz, \quad \alpha_{\text{shock}} = \frac{W_{R\phi}^{\text{shock}}}{\int P dz}. \quad (72)$$

The simulation should also provide the radial source functions $S_m(R, t)$ and $S_m l_{\text{in}}$ for the one-dimensional model.

12.2 Layer 2: vertical-equilibrium and S-curve solver

For a grid in (R, Σ) , solve equations (22), (23), and (36), including a chosen opacity, stress parameter μ , advection, and wind closure. The outputs are

$$T_c, \quad H, \quad \tau, \quad T_{\text{eff}}, \quad \dot{M}(\Sigma; R), \quad \Sigma_{\text{max}}(R), \quad \Sigma_{\text{min}}(R). \quad (73)$$

The first scan should vary

$$\alpha_c, \quad \alpha_h, \quad \mu, \quad \kappa, \quad Q_{\text{stream}}^+, \quad (74)$$

and determine whether the unstable radial interval lies within the tidally truncated minidisk.

12.3 Layer 3: one-dimensional time-dependent minidisk

Integrate equations (16)–(18) using the source functions from Layer 1 and thermal closure from Layer 2. The direct outputs are

$$\Sigma(R, t), \quad T_c(R, t), \quad \dot{M}_{\text{in}}(t), \quad \dot{M}_w(t), \quad \Delta M_{\text{cyc}}, \quad P, \quad D. \quad (75)$$

This calculation determines whether the transition is global or confined to a narrow annulus, whether cycles remain coherent under variable stream feeding, and whether the high-state duration is in the observed range.

12.4 Layer 4: wind and synthetic-emission model

The dynamical accretion rate must be converted into observable quantities:

$$L_X(t), \quad T_{\text{col}}(t), \quad R_{\text{ph}}(t), \quad L_{\text{UV}}(t). \quad (76)$$

A minimal treatment should include photon trapping, a radius-dependent wind mass-loss prescription, and reprocessing by an optically thick photosphere. The UV delay can then be tested against either light-travel or diffusion times.

13 Falsifiable conditions

The model is disfavored if any of the following occurs over the plausible parameter range:

1. captured gas does not circularize into a disk satisfying $R_{\text{ISCO}} < R_c < R_{\text{out}}$;
2. the local equilibrium curve is single-valued for realistic stress and magnetic support;
3. the unstable annulus lies outside the tidally allowed minidisk;
4. the participating mass is too small to supply the flare energy or too large to reload on the observed timescale;
5. the hot branch drains much faster than the observed multi-day flare duration;
6. the required average captured supply exceeds that available in a plausible transient TDE disk;
7. the wind/reprocessing model cannot produce the observed soft X-ray luminosity and UV response.

A successful model must satisfy the conservation-law checks

$$E_{\text{flare}} = \int L_{\text{bol}}(t) dt, \quad \Delta M_{\text{acc}} \geq \frac{E_{\text{flare}}}{\eta_{\text{max}} c^2}, \quad (77)$$

and

$$\langle \dot{M}_{\text{cap}} \rangle \geq \frac{\Delta M_{\text{acc}}}{P_{\text{QPE}}}. \quad (78)$$

14 Conclusions

The most defensible version of the IMRI-in-a-TDE-disk idea is not a density threshold for all gas inside the Hill sphere. It is a thermal–viscous state transition of a bound, stream-fed circumsecondary disk. The Hill flow sets the captured mass, angular momentum, and shock heating; the minidisk’s stress, opacity, advection, and winds determine whether an S-curve and limit cycle exist.

A simplified total-pressure α model gives a transparent criterion, $\mu < 4/7$, for a radiation-pressure thermal instability and yields a characteristic participating mass of order 10^{-5} – $10^{-4} M_{\odot}$ for fiducial IMRI parameters. With plausible supply and hot-state transport, the resulting loading and draining times can be comparable to the observed recurrence and burst durations, and the available accretion power can reach the Ansky regime. These numerical coincidences motivate further study but do not establish the model.

The highest-priority calculation is therefore a coupled program: three-dimensional hydro to determine Hill-sphere capture and shock transport, followed by a one-dimensional radiation-disk calculation to establish whether a tidally truncated minidisk actually possesses the required two stable branches and a nonlinear limit cycle.

A Useful normalized scalings

For the fiducial system,

$$R_H \simeq 1.10 \times 10^{12} \left(\frac{M_\bullet}{10^6 M_\odot} \right) \left(\frac{a}{50 r_g} \right) \left(\frac{q}{10^{-2}} \right)^{1/3} \text{ cm}, \quad (79)$$

$$\Omega_K(R_u) \simeq 6.05 \times 10^{-3} \left(\frac{M_2}{10^4 M_\odot} \right)^{1/2} \left(\frac{R_u}{3.31 \times 10^{11} \text{ cm}} \right)^{-3/2} \text{ s}^{-1}. \quad (80)$$

Equations (38)–(39) become

$$T_{\text{tr}} \simeq 2.08 \times 10^6 \left(\frac{\alpha_c}{0.01} \right)^{-1/4} \left(\frac{\kappa}{0.34 \text{ cm}^2 \text{ g}^{-1}} \right)^{-1/4} \left(\frac{\Omega_K}{6.05 \times 10^{-3} \text{ s}^{-1}} \right)^{1/4} \text{ K}, \quad (81)$$

$$\Sigma_{\text{tr}} \simeq 9.41 \times 10^5 \left(\frac{\alpha_c}{0.01} \right)^{-7/8} \left(\frac{\kappa}{0.34 \text{ cm}^2 \text{ g}^{-1}} \right)^{-7/8} \left(\frac{\Omega_K}{6.05 \times 10^{-3} \text{ s}^{-1}} \right)^{-1/8} \left(\frac{\mu}{0.62} \right)^{1/2} \text{ g cm}^{-2}. \quad (82)$$

The last factor follows from $\mathcal{R} \propto \mu^{-1}$.

B Recommended minimal parameter study

A first compact study could use:

Component	Parameters	Primary diagnostic
3D hydro	q , H/a , imposed β , inflow entropy	f_{cap} , j_{in} , α_{shock}
Vertical solver	α , μ , opacity, wind/advection	existence and width of S-curve
1D evolution	supply history, source radius, tidal torque	P , D , ΔM , coherence
Emission model	wind efficiency, photosphere, color factor	L_X , T_X , UV lag

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